

FINAL REPORT - CUSTOMER CHURN OF TELECOM INDUSTRY

Customer churn is a critical issue in the telecom industry which effect on their growth and profitability. Here the whole analysis is carried out by using two tools – python for data cleaning and exploratory data analysis and power bi for an interactive dashboard. This analysis helps to customer churn patterns, identify high risk segments and recommend actionable strategies for retention.

Key Analyst Questions

Before diving into findings, the analysis was guided by the following questions:

1. **What is the overall churn rate?**
2. **Which customer segments churn more?**
3. **Does tenure affect churn?**
4. **Which services impact churn?**
5. **What financial factors drive churn?**
6. **Which payment methods are riskier?**

Findings

✓ Numerical Insights

- Average tenure is 2.5 years but range up to 6 years.
- The monthly charges vary between 18 to 119 (in dollars).
- The total charges depend on tenure and monthly charges vary between 19 to 8685 (in dollars).

✓ Demographics

- 16% of customers are senior citizens.
- Majority is male.
- Most do not have dependents.

✓ **Services**

- Most common internet type: Fiber optic.
- Majority have phone service.
- Majority do not subscribe to extra services (security, backup, streaming, tech support).

✓ **Contracts & Billing**

- Most are on month-to-month contracts.
- Majority prefer paperless billing.
- Most common payment method: Electronic check.

Answers to Analyst Questions

1. Overall churn rate → 26.5% (27% customers left, 73% retained).
2. Demographic Analysis:
 - Senior citizens churn more (~42%).
 - Customers without partners churn ~33% vs ~19.7% with partners.
 - Customers without dependents churn ~31% vs ~15% with dependents.
 - Gender is not a significant driver (~26.2% for both).
3. **Tenure effect** → Long tenure customers are more loyal than new customers.
4. **Services impact** → Lack of internet, streaming, and tech support correlates with higher churn.
5. **Financial drivers** → High monthly charges and month-to-month contracts increase churn risk. Customers with two-year contracts showed the lowest churn, suggesting long-term commitments improve retention. One-year contracts fall in between, with churn higher than two-year but lower than month-to-month.
6. **Payment methods** → Electronic check is riskiest, auto-pay (bank transfer, credit card) is safer.

Suggestions

- Promote **long term contracts** (1–2 years).
- Offer **discounts** for high-charge customers.
- Encourage **auto-pay methods** (bank transfer, credit card).
- Bundle **extra services** (streaming, tech support, security).
- Target **senior citizens and single customers** with personalized offers.

Conclusion

The analysis reveals that churn is driven by contract type, payment method, monthly charges and lack of bundled services. By implementing strategies such as long-term contracts, auto-pay adoption and service bundles, telecom companies can reduce churn and strengthen customer loyalty.